

二維 Mindlin 板自由振動數值方法之深入分析報告

實驗分析組

June 24, 2026

參數	設定值
楊氏模量 (E)	1.0×10^6
泊松比 (ν)	0.3
板厚寬比 (h/b)	0.001
密度 (ρ)	1.0
網格切割份數 (n_{div})	$9 \rightarrow 25$

1 自由振動解析解

以下為該理論在四邊簡支 (SSSS) 邊界條件下，自由振動控制方程退化為代數矩陣的精確數學表達式：位移場與轉角場假設 (Navier Boundary Conditions)

$$w(x, y, t) = W_{mn} \sin(\alpha_m x) \sin(\beta_n y) e^{i\omega t}$$

$$\psi_x(x, y, t) = X_{mn} \cos(\alpha_m x) \sin(\beta_n y) e^{i\omega t}$$

$$\psi_y(x, y, t) = Y_{mn} \sin(\alpha_m x) \cos(\beta_n y) e^{i\omega t}$$

其中 $\alpha_m = \frac{m\pi}{a}$ ， $\beta_n = \frac{n\pi}{b}$ 。廣義特徵值矩陣方程 (Generalized Eigenvalue Problem)

$$\begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix} \begin{Bmatrix} W_{mn} \\ X_{mn} \\ Y_{mn} \end{Bmatrix} = \omega^2 \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & 0 \\ 0 & 0 & M_{33} \end{bmatrix} \begin{Bmatrix} W_{mn} \\ X_{mn} \\ Y_{mn} \end{Bmatrix}$$

剛度矩陣元 (Stiffness Matrix Elements)

$$K_{11} = D^s(\alpha_m^2 + \beta_n^2)$$

$$K_{12} = D^s \alpha_m$$

$$K_{13} = D^s \beta_n$$

$$K_{21} = D^s \alpha_m$$

$$K_{22} = D\alpha_m^2 + D\left(\frac{1-\nu}{2}\right)\beta_n^2 + D^s$$

$$K_{23} = D\left(\frac{1+\nu}{2}\right)\alpha_m\beta_n$$

$$K_{31} = D^s \beta_n$$

$$K_{32} = D \left(\frac{1+\nu}{2} \right) \alpha_m \beta_n$$

$$K_{33} = D \beta_n^2 + D \left(\frac{1-\nu}{2} \right) \alpha_m^2 + D^s$$

質量矩陣元 (Mass Matrix Elements)

$$M_{11} = I_0 = \rho h$$

$$M_{22} = I_2 = \frac{\rho h^3}{12}$$

$$M_{33} = I_2 = \frac{\rho h^3}{12}$$

材料與幾何常數定義

- 彎曲剛度： $D = \frac{Eh^3}{12(1-\nu^2)}$
- 剪切剛度： $D^s = \kappa^2 Gh = \frac{5}{6} \frac{E}{2(1+\nu)} h$

```

1 function exact_omega_sq(m, n, a, b, Db, ρ, h)
2     v_ex = 0.3
3     E_ex = Db * 12 * (1 - v_ex^2) / h^3
4     Ds_ex = (5/6) * E_ex * h / (2 * (1 + v_ex))
5
6     α_m = m * π / a; β_n = n * π / b; λ^2 = α_m^2 + β_n^2
7
8     K_ex = zeros(3,3)
9     K_ex[1,1] = Ds_ex * λ^2; K_ex[1,2] = Ds_ex * α_m; K_ex[1,3] = Ds_ex * β_n
10    K_ex[2,1] = Ds_ex * α_m; K_ex[2,2] = Db * α_m^2 + Db * ((1-v_ex)/2) * β_n^2 + Ds_ex; K_ex[2,3] = Db * ((1+v_ex)/2) * α_m * β_n
11    K_ex[3,1] = Ds_ex * β_n; K_ex[3,2] = K_ex[2,3]; K_ex[3,3] = Db * β_n^2 + Db * ((1-v_ex)/2) * α_m^2 + Ds_ex
12
13    M_ex = zeros(3,3)
14    M_ex[1,1] = ρ * h; M_ex[2,2] = ρ * h^3 / 12; M_ex[3,3] = ρ * h^3 / 12
15
16    vals = eigvals(K_ex, M_ex)
17    ω_exact = sqrt(minimum(real(vals)))
18    w_h_exact = ω_exact * a^2 * sqrt(ρ * h / Db)
19    return w_h_exact, w_h_exact
20 end

```

2 網格

Vibration_Mix 和縮減積分：

```

1 mesh_files = [normpath(joinpath(@__DIR__, "..", "msh", "st_q_$i.msh")) for i in 9:25]

```

Vibration_mixtwo：

3 誤差收斂分析

```

1  for n_div in ndiv_series
2      ndiv_w = n_div - 2; ndiv_phi = n_div; ndiv = n_div
3      mesh_file_w = normpath(joinpath(@_DIR_, "..", "msh", "st_q_$(ndiv_w).msh"))
4      mesh_file_phi = normpath(joinpath(@_DIR_, "..", "msh", "st_q_$(ndiv_phi).msh"))
5      mesh_file_s = normpath(joinpath(@_DIR_, "..", "msh", "st_q_$(ndiv).msh"))

```

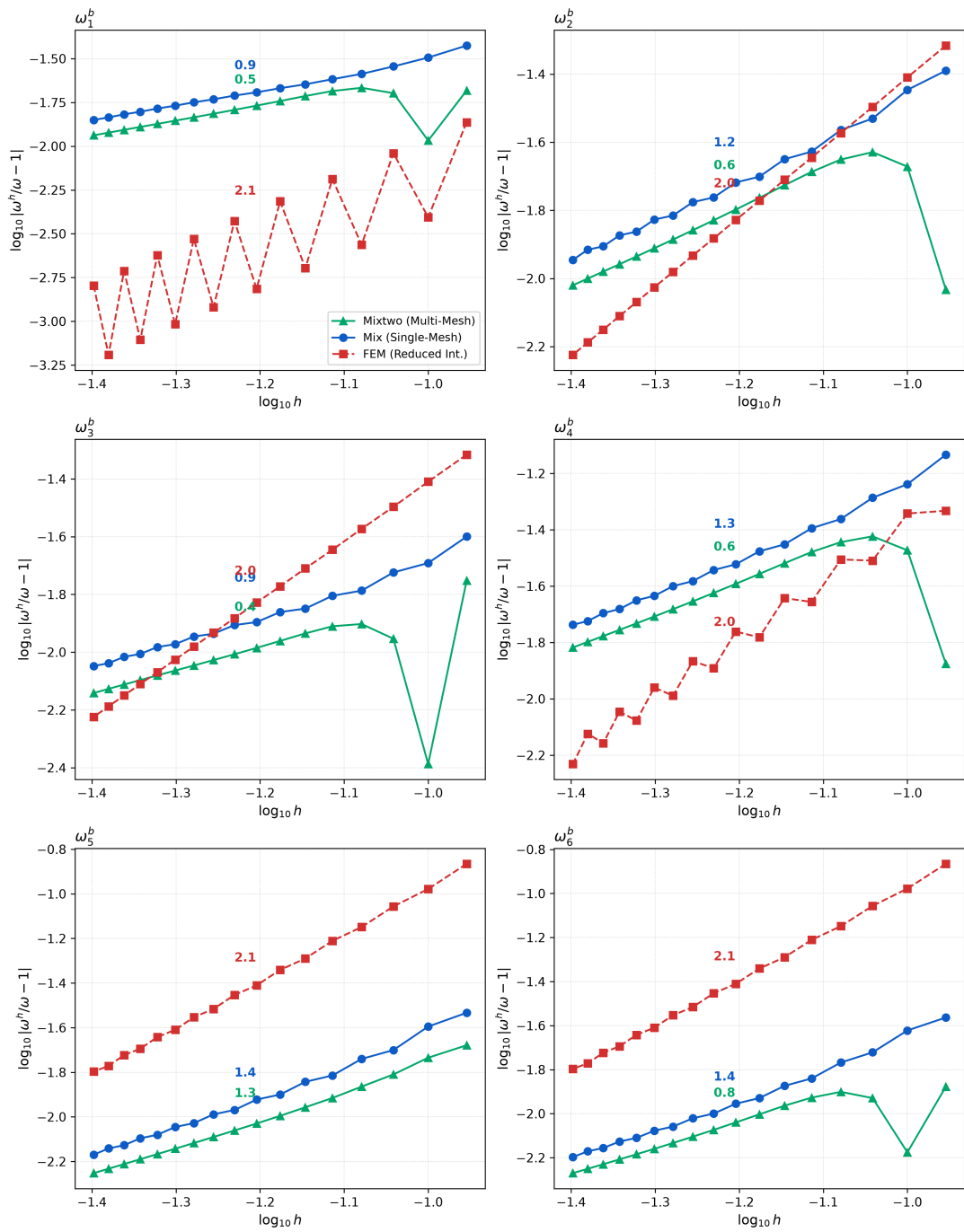


Figure 1: 前 6 階模態之正規化頻率分佈